Lab 8 Answers:

1. The audio element is stored in `useRef` because it is a persistent browser object that needs to be controlled through methods such as `play()`, `pause()`, and `currentTime`. Changing a ref does not trigger a React render, while state is used separately for information the interface needs to display, such as playback status and progress.

2. The service files separate data-fetching logic from the user interface. This makes the application easier to maintain because components can reuse the same functions without duplicating API URLs, request handling, and response processing.

3. React code is delivered to the browser, where users can inspect the bundled JavaScript and potentially discover an embedded API key. If the key is exposed on GitHub, someone else could use it to make requests, consume the available quota, or incur charges depending on the account's configuration; therefore, the key should be stored in a server-side environment variable.

4. The recommender is an LLM prompt because it sends the user's saved albums and ratings to an already-trained language model and asks it to generate suggestions, rather than training a new model on a dataset. One strength is that it can use descriptions of musical styles and patterns to suggest diverse albums, while one weakness is that it can produce inaccurate suggestions or incorrect reasons.

5. The hardest bugs were imports and finding the wrong or misplaced characters.